Testing Efficacy of Mass-Radius Relations Using Exoplanet GJ 436 b

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1 Motivation

Understanding the fundamental properties of celestial bodies is essential to being able to draw useful conclusions about their formation, lifespan, and more. However, observation methods used to discover new planets often lack the ability to determine all physical properties of the body; instead, these methods are typically confined to defining mass or radius, but never both at once. Because of this, there is a need to find a relationship between these properties in order to forecast one using the other.

By analyzing transit and radial velocity data associated with one particular exoplanet (GJ 436 b), we can test proposed relationships between an exoplanet's mass and radius. This will be accomplished by performing three separate sets of calculations: (1) determining mass using radial velocity data, (2) determining radius using transit data, and (3) relating the mass and radius using a density calculation. These results will then be compared against the mass-radius relationship proposed in Chen & Kipping (2016).

2 Methods

2.1 Radial Velocity

In the search for exoplanets, the radial velocity method is largely considered the most effective form of exoplanet detection. By measuring shifts in the wavelength of the light emanated by the star, it is possible to detect periodic, radial motion that implies the existence of an exoplanet. When approximated to first order, the semi-amplitude K of the radial velocity gives the planet's mass, which can be calculated using the following equation:

$$\mathrm{K} = \frac{M_\mathrm{p}}{M_*}\sqrt{\frac{GM_*}{a}} * \sin(i)$$

We can rearrange this equation to solve for the mass of the planet:

$$M_\mathrm{p} = \frac{KM_*}{\sin(i)}\sqrt{\frac{a}{GM_*}}$$

Using values K = 17.09±0.22 m/s, M_star = 0.441±0.00938M_sun, inclination i=86.44+0.17-0.16 deg, semi-major axis a=0.0291±0.0015 AU [3], and the gravitational constant G we can use this formula to calculate the mass of GJ 436 b.

2.2 Transit

If an exoplanet's orbit happens to cross directly between the host star and the observer, i.e., it is on the same plane of inclination, then we will observe a brief periodic dip in the luminosity of the host star. This dip is the planet obscuring light during its "transit" in front of the star. The amount of light the planet blocks, also known as the transit depth, reflects the planet size relative to the host star.

To properly generate a model for our transit, we first generated values using the EXOFAST program offered on NASA's exoplanet archive [3]. The inputs used planetary and stellar data supplied by Maciejewski et al. 2014 [6], whereas stellar temperature and metallicity data was supplied by Rosenthal et al. 2021[7]. EXOFAST provides the limb darkening coefficients, eccentricity, planet radius over stellar radius, period, semi-major axis to stellar radius ratio, inclination, periastron, and the mid time for our transit.

We use the pylightcurve.transit() function to generate a flux array to model our transit. This was done using the data sourced from Demming et al. 2007[4], with both our model and original data being visualized below in Figure 1.

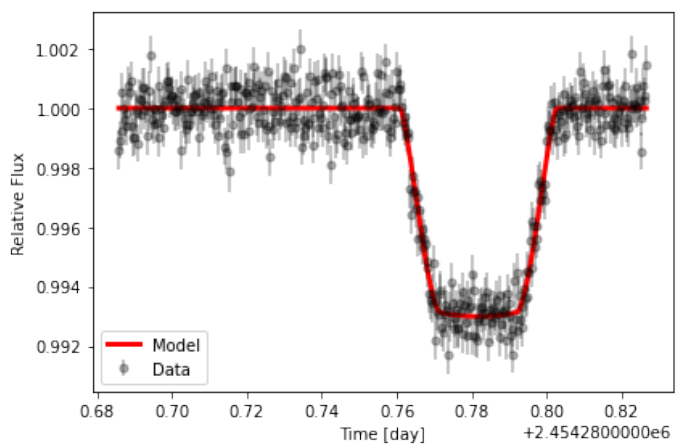

Figure 1: Transit of GJ 436 b

From this we could observe a distinct dip in the data. This depth of transit is represented by the value of δ. We identify this value by finding the difference in the peak and minimum value of our modeled flux curve. Additionally, this value of δ has a degree of uncertainty; found by calculating the standard deviation of the points located within the bottom of the dip. It was found that the depth of the transit is $\delta = 7.003e-3 \pm 5.9e-4$. After obtaining this value of δ we can then calculate the radius of GJ 436 b using the formula:

$$\delta = \left(\frac{R_{\mathrm{p}}}{R_{\mathrm{star}}}\right)^2$$

where R_p is the radius of our GJ 436 b and R_{star} is the radius of the star GJ 436, sourced from Rosenthal et al. 2021[7].

2.3 Calculating Density

Once we have found the mass and radius of a planet, we can solve for its density. If we assume that an exoplanet has a perfectly-spherical shape, then we can calculate its average density $\overline{\rho}$ as

$$\overline{\rho} = \frac{m_{\text{exo}}}{V_{\text{sphere}}} = \frac{3m_{\text{exo}}}{4\pi r_{\text{exo}}^3}$$

Using the values we found in section 3.1 and 3.2, we apply this formula to calculate the average density of GJ 436 b. Note that the spherical assumption made for this calculation is not entirely accurate. Depending on an exoplanet's composition, its shape could be more jagged and predictable. However, since GJ 436 b is a gas giant rather than a terrestrial planet, the exoplanet's shape is more likely to be spherical. Hence, the spherical approximation is accurate enough for our purposes.

2.4 Error Propagation

While the above sections discuss how to calculate the mass, radius, and density of an exoplanet, they do not account for how uncertainty propagates through these equation. In order to understand the error that results from these calculations, we refer to the following equations. For some general quantity Q given by

$$Q = \frac{a, b, ...c}{x, y, ...z}$$

where the uncertainty in each number is represented as δ(number), we have that

$$\frac{\delta Q}{|Q|} = \sqrt{\left(\frac{\delta a}{a}\right)^2 + \left(\frac{\delta b}{b}\right)^2 + ... + \left(\frac{\delta c}{c}\right)^2 + \left(\frac{\delta x}{x}\right)^2 + \left(\frac{\delta y}{y}\right)^2 + ... + \left(\frac{\delta z}{z}\right)^2}$$

This allows us to calculate the uncertainty of the ratio between two values. We must also understand how raising a number to some power n will affect uncertainty. Using that $Q = x^n$, we have that the uncertainty in Q is given by

$$\frac{\delta Q}{|Q|} = |n| \frac{\delta x}{|x|}$$

These two equations are vital in calculating the uncertainty in an exoplanet's mass and radius, since the calculations involve division and powers. However, the mass calculation involved a trig term, and the inclination we used has an associated uncertainty, so we must define how trig terms affect uncertainty. Given some inclination $i = a \pm b$ the uncertainty in $Q = \sin i$

$$\delta Q = |\sin(a) - \sin(a + b)|$$

With these results, we can finally calculated the uncertainty associated with an exoplanet's mass, radius, and density.

3 Results

3.1 Mass, Radius, & Density Comparison

GJ 436 b is a Neptune-sized exoplanet that orbits an M-type star. Our results estimated a mass of 22.1±2.3 M_{earth} and a radius of 0.326 ± 0.025 R_{Jupiter}. Using these results, we found that exoplanet GJ 436 b has a density of $2.6 + -0.2 g/cm^3$. Since GJ 436 b has a confirmed mass of 22.1±2.3 $M_{\oplus}$, and radius of 0.372±0.015 R_{Jupiter}[2], our values are within two standard deviations of the accepted values. In order to compare our results to similar exoplanets, we source data for all known exoplanets from NASA's exoplanet archive. Since density is not provided in NASA's exoplanet data, we manually calculated using the reported exoplanet masses and radii. This means that any exoplanets without a reported mass or radius were not considered in our study.

Planet Name	M_{exo} ($M_{\oplus}$)	R_{exo} (R_{Jupiter})	ρ_{exo} (g/cm^3)
2MASS J02192210-3925225 b	1.440	4417.83700	6.173293
2MASS J21402931+1625183 A b	0.920	6657.48000	35.673115
55 Cnc e	0.167	7.99000	7.158005
AU Mic b	0.363	17.00000	1.482940
AU Mic c	0.289	13.60000	2.350932
BD+20 594 b	0.230	22.24810	7.629640
BD+60 1417 b	1.310	4767.42610	8.848400
CFHTWIR-Oph 98 b	1.860	2479.06157	1.607471
COCONUTS-2 b	1.110	2002.31896	6.108843
CT Cha b	2.200	5403.00000	2.117198

These are convenient values, since they are directly comparable to the mass and radius of GJ 436 b. Before we can plot these, we limit the scope of our data to exoplanets with similar masses and radii to GJ 436 b. Our group determined that radii and masses that differ by 50% was sufficient. We can now compare our exoplanet (GJ 436 b) to similar-sized exoplanets, as shown in Figure 2 below.

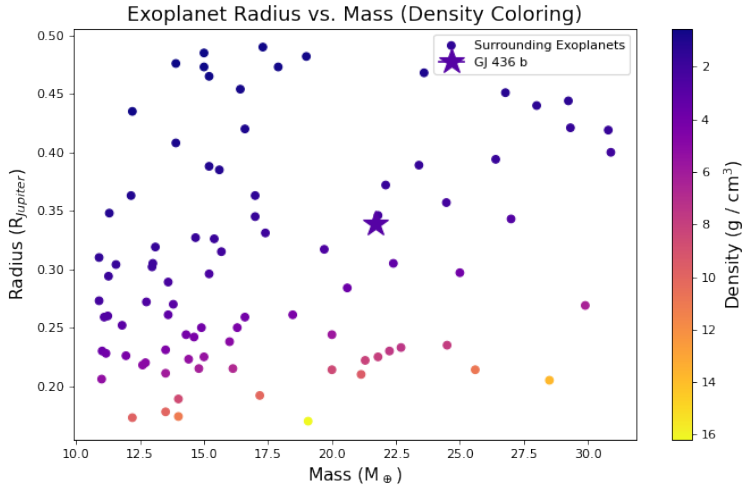

Figure 2: Comparison of GJ 436 b density to exoplanets of similar size.

From this figure, we can see that our results are reasonable in comparison to other exoplanets of similar size, as our exoplanet (represented by the star) fits in well amongst the other data. Additionally, our calculated density matches that stipulated by the density gradient of the other exoplanets, again indicating a good fit.

3.2 Mass-Ratio Relation Comparison

In their 2016 paper, Chen & Kipping proposed numerous different power laws used to predict the radius of an exoplanet as a function of mass. These predictions were segregated into four categories: Terran Worlds, Neptunian Worlds, Jovian Worlds, Stellar World. The distinction between Terran and Neptunian worlds is approximately $2R_{\text{Earth}}$, indicating that GJ 436 b is, as expected, approximately Neptune-sized. Thus, Chen & Kipping predicts that GJ 436 b would follow a ratio of $R/R_{\text{Earth}} \approx (M/M_{\text{Earth}})^{0.59}$, resulting in a predicted radius of 0.561±0.024 R_{Jupiter} given our calculated mass value. However, the radius we calculated is 0.339±0.015 R_{Jupiter}. It is important to note that this value is within two standard deviations of the radius produced.

4 Conclusion

Overall, our calculations of GJ 436 b had a high degree of accuracy when compared to the NASA Exoplanet Archive results. Discrepancies in our planetary radii compared to the Maciejewski et al. 2014 paper can be traced to a difference in the value used for the stellar radius. Using the same stellar radius that Glen 2014 used produced identical results, however, using more recently published data produced a planetary radius several standard deviations away from the conclusions of Maciejewski et al. 2014. This could warrant further investigation of the planetary radius of GJ 436 b using our current knowledge regarding the host star.

Additionally, when our results are compared to the results of Chen & Kipping, we find that the two do not precisely match. We achieved a radius of 0.339±0.015 R_{Jupiter}, and this measured uncertainty does not put us within the bounds of the results of Chen & Kipping, even if we are to assume maximum error for both results. As such, it cannot be said that our results match that of Chen & Kipping, though they are within an order of magnitude.

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